A Tableau Calculus for $\mathcal{ALCI}_{RCC5}$ with Tri-Neighborhood Blocking

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Abstract

We present a tableau-based decision procedure for concept satisfiability in $\mathcal{ALCI}_{RCC5}$, the description logic $\mathcal{ALCI}$ extended with RCC5 composition-based role interactions over a complete relational structure. The key innovation is *Tri-neighborhood blocking*: a node x is blocked by an earlier node y only when they share the same concept label, the same set of abstract triangle types—tuples $(\tau_1, R_{12}, \tau_2, R_{23}, \tau_3, R_{13})$ recording Hintikka types and RCC5 relations but no node identities—and, for each (relation, type) pair, the same distribution of triangle-type sets among their respective neighbors. We prove termination (the neighborhood signature is drawn from a finite universe), soundness (the model is constructed by triangle-type-filtered unraveling, with global consistency guaranteed by the full tractability of RCC5), and completeness (a model guides all nondeterministic choices). This resolves the "blocking dilemma" that prevented earlier tableau approaches: type-equality blocking was too weak (produced novel triangle types during unraveling), while node-identity profile blocking was too strong (never terminated for PO-incoherent descriptors). Tri-neighborhood blocking achieves both properties simultaneously, and the strengthened neighborhood condition ensures that the copy is faithful from every participating node's perspective.

1 Introduction

The description logic $\mathcal{ALCI}_{RCC5}$ extends $\mathcal{ALCI}$ with a role box defined by the RCC5 composition table: the five base relations $\{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ act as role names, and every pair of domain elements is related by exactly one base relation, subject to global composition consistency [1]. Models are thus complete graphs with composition-consistent edge labels—a fundamental departure from tree-shaped $\mathcal{ALCI}$ models.

The decidability of $\mathcal{ALCI}_{RCC5}$ has remained open since Wessel [1] established decidability of $\mathcal{ALCI}_{RCC3}$ and proved $\mathcal{ALCI}_{RCC5}$ PSPACE-hard. Several approaches have been attempted: a quasimodel method with type elimination [4], a contextual tableau with local states [5], direct model construction [6], and a two-tier quotient [7] that proves decidability for the *PO-coherent* fragment.

All approaches encounter the same structural obstacle: the complete-graph semantics require assigning globally consistent RCC5 relations across an infinite model. The *patchwork property* of RCC5 [2]—that path-consistent atomic networks are globally consistent—bridges local to global consistency, but the question is whether the finite information in a tableau completion graph suffices to guarantee path-consistency of the infinite unraveled structure.

The blocking dilemma

In a standard DL tableau, blocking ensures termination: a blocked node x copies the witness structure of its blocker y during model extraction. In complete-graph models, this copying ("unraveling") must preserve RCC5 composition consistency across *all* triples, not just the parent-child edge.

This creates a dilemma, identified in [7]:

- **Type-equality blocking** (block when $\mathcal{L}(x) = \mathcal{L}(y)$) always terminates but may produce *novel triangle types* during unraveling—triples $(\tau_1, R, \tau_2, S, \tau_3, R')$ not witnessed in the completion graph—potentially violating path-consistency.
- **Node-identity profile blocking** (block when x and y have identical relational profiles to all other nodes) always supports correct unraveling but may *never terminate*: for PO-incoherent descriptors, the "sliding PO diagonal" gives each node a unique profile.

Tri-neighborhood blocking

We resolve this dilemma by observing that the non-termination argument uses *concrete* node identities, while the correctness argument only needs *abstract* relational patterns. An *abstract triangle type* is a tuple $(\tau_1, R_{12}, \tau_2, R_{23}, \tau_3, R_{13})$ recording three Hintikka types and three pairwise RCC5 relations—no node identities. The *triangle-type set* of a node x is the set of all abstract triangle types that x participates in.

Computational verification on the PO-incoherent counterexample (a 24-element PP-chain with 12 PO-witnesses) shows that abstract triangle-type sets *stabilize*: all interior τ_A nodes share exactly the same 68 abstract triangle types, despite having different concrete neighbors. The stabilization occurs because the abstract triangle-type set depends on the *menu* of available (type, relation) pairs among neighbors, not on the *count* of neighbors of each kind.

We strengthen this to *Tri-neighborhood blocking*: in addition to requiring $\mathcal{L}(x) = \mathcal{L}(y)$ and $\mathsf{Tri}(x) = \mathsf{Tri}(y)$, we require that for each (relation, type) pair, the set of Tri-values among the respective neighbors of x and y also matches. This ensures the copy is faithful not just from x/y's perspective, but from *every neighboring node's perspective*. Computational verification confirms that this stronger condition also stabilizes (at depth $k = 3$, one step later than basic Tri-matching at $k = 2$).

Tri-neighborhood blocking achieves:

1. **Termination**: the neighborhood signature is drawn from a finite universe, and stabilization ensures matching.
2. **Correct unraveling**: copying y's witness structure for x produces only triangle types already in the completion graph's set T, verified from every participating node's perspective.
3. **Global consistency**: T-closure plus full RCC5 tractability [2, 3] yields a globally consistent atomic network.

2 Preliminaries

2.1 The logic $\mathcal{ALCI}_{RCC5}$

Concepts are built from concept names, the RCC5 base relations $R \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ as role names, and the constructors $\sqcap$, $\sqcup$, $\neg$, $\exists R.C$, $\forall R.C$ in negation normal form (NNF). We adopt **strong EQ semantics**: $\rho(d, e) = \mathsf{EQ}$ iff $d = e$.

An interpretation $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$ assigns a total function $\rho^{\mathcal{I}} : \Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}} \to \{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ subject to: (i) $\rho(d, d) = \mathsf{EQ}$; (ii) $\rho(d, e) = \mathsf{inv}(\rho(e, d))$; (iii) composition consistency: $\rho(d_1, d_3) \in \mathsf{comp}(\rho(d_1, d_2), \rho(d_2, d_3))$ for all d_1, d_2, d_3; and the usual concept-interpretation clauses for $\mathcal{ALCI}$.

2.2 Types, closure, composition table

Given C_0 in NNF, $\mathsf{cl}(C_0)$ is the Fischer–Ladner closure. A *Hintikka type* $\tau \subseteq \mathsf{cl}(C_0)$ is a maximal propositionally consistent subset. Write $\mathsf{Tp}(C_0)$ for the set of types; $|\mathsf{Tp}(C_0)| \leq 2^{|\mathsf{cl}(C_0)|}$.

Write $\mathsf{NR}^- = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$ for the base relations between distinct elements (EQ excluded).

The composition table $\mathsf{comp} : (\mathsf{NR}^-)^2 \to 2^{\mathsf{NR}^-}$ is:

$\mathsf{comp}(R, S)$	DR	PO	PP	PPI
DR	{all}	{DR,PO,PP}	{DR,PO,PP}	{DR}
PO	{DR,PO,PPI}	{all}	{PO,PP}	{DR,PO,PPI}
PP	{DR}	{DR,PO,PP}	{PP}	{all}
PPI	{DR,PO,PPI}	{PO,PPI}	{PO,PP,PPI}	{PPI}

2.3 Type-safety

Definition 2.1 (Type-safe relations)**.** For Hintikka types τ_1, τ_2, define

$$\mathsf{Safe}(\tau_1, \tau_2) = \big\{R \in \mathsf{NR}^- : \forall\,(\forall R.C) \in \tau_1,\ C \in \tau_2;\ \forall\,(\forall \mathsf{inv}(R).C) \in \tau_2,\ C \in \tau_1\big\}.$$

In any model, $\rho(d, e) \in \mathsf{Safe}(\mathsf{tp}(d), \mathsf{tp}(e))$ for distinct d, e.

2.4 Full tractability of RCC5

Theorem 2.2 (Renz & Nebel [2], Renz [3])**.** *Every path-consistent disjunctive RCC5 constraint network over* $\{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ *is satisfiable. Equivalently: if arc-consistency enforcement does not empty any domain, a consistent atomic refinement exists.* □

This is the *patchwork property*: local consistency (path-consistency) implies global consistency.

3 The Tableau Calculus

3.1 Completion graphs

Definition 3.1 (Completion graph)**.** A *completion graph* for C_0 is a tuple $\mathcal{G} = (V, \mathcal{L}, \mathcal{E}, \prec)$ where:

- V is a finite non-empty set of nodes.
- $\mathcal{L} : V \to 2^{\mathsf{cl}(C_0)}$ is a labelling function.
- $\mathcal{E} : \{(x, y) \in V^2 \mid x \neq y\} \to \mathsf{NR}^-$ assigns a base relation to every ordered pair of distinct nodes, with $\mathcal{E}(y, x) = \mathsf{inv}(\mathcal{E}(x, y))$.
- $\prec$ is a strict total order on V (creation order).

The *initial* completion graph has a single node x_0 with $\mathcal{L}(x_0) = \{C_0\}$.

3.2 Triangle types

Definition 3.2 (Abstract triangle type)**.** An *abstract triangle type* is a tuple $(\tau_1, R_{12}, \tau_2, R_{23}, \tau_3, R_{13})$ where $\tau_1, \tau_2, \tau_3 \in \mathsf{Tp}(C_0)$ and $R_{12}, R_{23}, R_{13} \in \mathsf{NR}^-$ with $R_{13} \in \mathsf{comp}(R_{12}, R_{23})$.

Definition 3.3 (Triangle-type set of a node)**.** For a node x in $\mathcal{G}$, the *triangle-type set* of x is:

$$\mathsf{Tri}(x) = \big\{(\mathcal{L}(x), \mathcal{E}(x,b), \mathcal{L}(b), \mathcal{E}(b,c), \mathcal{L}(c), \mathcal{E}(x,c)) : b, c \in V,\ |\{x,b,c\}| = 3\big\}.$$

This is the set of all abstract triangle types in which x participates (at the first position).

Remark 3.4. Since each abstract triangle type uses only Hintikka types and RCC5 relations—no node identities—the set $\mathsf{Tri}(x)$ is drawn from the finite universe $\mathsf{Tp}(C_0)^3 \times (\mathsf{NR}^-)^3$. Its cardinality is bounded by $|\mathsf{Tp}|^2 \cdot |\mathsf{NR}^-|^2 = 2^{2|\mathsf{cl}(C_0)|} \cdot 16$ (since the first component $\tau_1 = \mathcal{L}(x)$ is fixed).

Definition 3.5 (Triangle-type set of the graph)**.** The *triangle-type set* of $\mathcal{G}$ is:

$$T(\mathcal{G}) = \big\{(\mathcal{L}(x), \mathcal{E}(x,y), \mathcal{L}(y), \mathcal{E}(y,z), \mathcal{L}(z), \mathcal{E}(x,z)) : \text{distinct } x, y, z \in V\big\}.$$

3.3 Tri-neighborhood signature

Definition 3.6 (Tri-neighborhood signature)**.** For a node x in $\mathcal{G}$, the *Tri-neighborhood signature* of x is:

$$\mathsf{TNbr}(x) = \big\{\big((R,\tau),\ \{\mathsf{Tri}(b) : b \in V,\ b \neq x,\ \mathcal{E}(x,b) = R,\ \mathcal{L}(b) = \tau\}\big) : R \in \mathsf{NR}^-,\ \tau \in \mathsf{Tp}\big\}.$$

That is, for each (relation, type) pair (R, τ), $\mathsf{TNbr}(x)$ records the *set* of distinct Tri-values occurring among x's R-neighbors of type τ.

Remark 3.7. The Tri-neighborhood signature captures not only which abstract triangle types x participates in, but also which abstract triangle types each of x's neighbors participates in, grouped by how the neighbor relates to x. This is analogous to *pairwise blocking* in $\mathcal{ALCI}$ tree tableaux (which matches a node *and* its parent with the blocker and its parent), generalized to the complete-graph setting: instead of matching one parent edge, we match *all* neighbor edges grouped by (relation, type).

3.4 Blocking

Definition 3.8 (Tri-neighborhood blocking)**.** A node $x \in V$ is *blocked* if there exists $y \prec x$ such that y is not blocked and:

(i) $\mathcal{L}(x) = \mathcal{L}(y)$ (same concept label),
(ii) $\mathsf{Tri}(x) = \mathsf{Tri}(y)$ (same triangle-type set), and
(iii) $\mathsf{TNbr}(x) = \mathsf{TNbr}(y)$ (same Tri-neighborhood signature).

We call y a *blocker* for x. A node is *active* if it is not blocked.

Remark 3.9. Conditions (ii) and (iii) are the key innovations. Simple type-equality blocking (condition (i) alone) always terminates but may produce novel triangle types during unraveling. Condition (ii) ensures that x and y participate in the same abstract relational patterns. Condition (iii) additionally ensures that the *neighbors* of x and y also have matching triangle-type sets: for each (relation, type) pair, the menu of Tri-values among neighbors is identical.

This strengthening is crucial for the soundness proof: when we copy y's witness structure for blocked node x, we need the triangle types to be in T not just from x/y's perspective, but from *every neighbor's perspective*. Condition (iii) guarantees this.

Note that conditions (ii)–(iii) do *not* require x and y to be related to the same concrete nodes. They require only that the *abstract neighborhood patterns* match. This is strictly weaker than node-identity profile matching, which is why the condition still terminates (Section 4).

3.5 Expansion rules

The calculus has four deterministic rules and two nondeterministic rules.

Propositional rules (deterministic):

($\sqcap$) If $D_1 \sqcap D_2 \in \mathcal{L}(x)$ and $\{D_1, D_2\} \not\subseteq \mathcal{L}(x)$: add D_1 and D_2 to $\mathcal{L}(x)$.

($\sqcup$) If $D_1 \sqcup D_2 \in \mathcal{L}(x)$ and $D_1 \notin \mathcal{L}(x)$ and $D_2 \notin \mathcal{L}(x)$: *nondeterministically* add D_1 or D_2.

Propagation rule (deterministic):

($\forall$) If $\forall R.D \in \mathcal{L}(x)$ and $\mathcal{E}(x, y) = R$ and $D \notin \mathcal{L}(y)$: add D to $\mathcal{L}(y)$.

This handles inverses: $\forall R^-.D$ becomes $\forall \mathsf{inv}(R).D$ in NNF.

Generating rule (nondeterministic):

($\exists$) If $\exists R.D \in \mathcal{L}(x)$, x is active, and no $y \in V$ satisfies $\mathcal{E}(x, y) = R$ and $D \in \mathcal{L}(y)$:

1. Create a fresh node y, $\prec$-greatest.
2. Set $\mathcal{L}(y) := \{D\}$.
3. Set $\mathcal{E}(x, y) := R$, $\mathcal{E}(y, x) := \mathsf{inv}(R)$.
4. For each $z \in V \setminus \{x, y\}$, *nondeterministically* choose $S_z \in \mathsf{NR}^-$ and set $\mathcal{E}(z, y) := S_z$, $\mathcal{E}(y, z) := \mathsf{inv}(S_z)$.

3.6 Constraint filtering

After step 4 of ($\exists$), we require:

(CF) Composition consistency. For every triple of distinct nodes (u, v, w) involving y:

$$\mathcal{E}(u, w) \in \mathsf{comp}(\mathcal{E}(u, v), \mathcal{E}(v, w)).$$

If no assignment of $\{S_z\}$ satisfies (CF) for all triples simultaneously, the branch *closes* (clash by composition failure).

Remark 3.10. Checking (CF) is an RCC5 constraint satisfaction problem. By Theorem 2.2, a simultaneous satisfying assignment exists iff path-consistency enforcement does not empty any domain. This can be checked in $O(|V|^3)$.

3.7 Rule application strategy

1. Apply ($\sqcap$), ($\sqcup$), ($\forall$) exhaustively to all nodes.
2. Evaluate blocking: x is blocked if $\mathcal{L}(x) = \mathcal{L}(y)$, $\mathsf{Tri}(x) = \mathsf{Tri}(y)$, and $\mathsf{TNbr}(x) = \mathsf{TNbr}(y)$ for some non-blocked $y \prec x$.
3. Apply one instance of ($\exists$) for an active node with an unsatisfied demand.
4. Apply ($\forall$) exhaustively (new edges may trigger propagation).
5. Re-evaluate blocking.
6. Repeat from step 1 until no rule applies or a clash is detected.

3.8 Clash conditions

Definition 3.11 (Saturated and clash-free)**.** $\mathcal{G}$ is *saturated* if no rule applies. $\mathcal{G}$ is *clash-free* if:

(C1) No x has $\{A, \neg A\} \subseteq \mathcal{L}(x)$.
(C2) Composition consistency (CF) holds for all triples.

A saturated, clash-free graph is *open*.

4 Termination

Theorem 4.1 (Termination)**.** *Every sequence of rule applications terminates.*

Proof. Let $n = |\mathsf{cl}(C_0)|$ and $N_\tau = |\mathsf{Tp}(C_0)| \leq 2^n$.

Finiteness of blocking signatures. An abstract triangle type is an element of $\mathsf{Tp}^3 \times (\mathsf{NR}^-)^3$. Let S denote the number of possible Tri-values; $S \leq 2^{N_\tau^3 \cdot 64}$. A Tri-neighborhood signature assigns, to each of the at most $4 \cdot N_\tau$ pairs (R, τ), a subset of $\{1, \ldots, S\}$ (a set of Tri-values). The number of distinct TNbr-values is at most $(2^S)^{4N_\tau}$. The number of distinct $(\mathcal{L}, \mathsf{Tri}, \mathsf{TNbr})$ triples is therefore bounded by $A(C_0) := N_\tau \cdot S \cdot (2^S)^{4N_\tau}$.

Active node bound. Active nodes have pairwise distinct $(\mathcal{L}, \mathsf{Tri}, \mathsf{TNbr})$ triples (by the blocking definition). Hence the number of *simultaneously* active nodes is at most $A(C_0)$.

Labels grow monotonically. Rules only add concepts to labels, never remove them. Each label changes at most n times. Edge labels $\mathcal{E}(x, y)$ are set at node creation and never change.

Bounded branching. The $(\exists)$-rule for node x and demand $\exists R.D$ fires only when no witness exists. Once a witness y is created (with $\mathcal{E}(x, y) = R$, $D \in \mathcal{L}(y)$), the demand is *permanently* satisfied: D remains in $\mathcal{L}(y)$ (labels only grow) and $\mathcal{E}(x, y)$ never changes. Therefore each node fires $(\exists)$ at most n times, creating at most n children in the creation tree.

Bounded total node creation. We must bound not just simultaneous active nodes but the *total* number of nodes ever created. (An earlier version of this proof appealed only to the active node bound, which is insufficient: a process with at most $A(C_0)$ active nodes could in principle cycle through blocking and unblocking, creating fresh successors each time.)

The correct argument has two parts.

Part 1: Permanent demand satisfaction limits re-expansion. When a previously blocked node x becomes unblocked (because its blocker's signature changed), x may fire $(\exists)$ for its remaining unsatisfied demands. But demands satisfied in earlier active periods remain satisfied (labels and edges are permanent). So x fires $(\exists)$ at most n times *total*, regardless of how many blocking/unblocking cycles it undergoes.

Part 2: Blocking/unblocking cycles are bounded per node. A node x transitions from blocked to active only when its own signature or its blocker's signature changes. Each such change requires either (a) a label growth at some node (at most n per node, irreversible), or (b) a $\mathsf{Tri}/\mathsf{TNbr}$ change caused by a new node creation or a label growth at a neighbor. In the *label-stable regime* (after all labels have reached their final values), $\mathsf{Tri}(x)$ can only grow monotonically: new nodes may add new triangle types, but no types are replaced (since the Hintikka-type component of each triangle is now fixed). Similarly, $\mathsf{TNbr}(x)$ can only grow. Since $\mathsf{Tri}(x) \subseteq T_{\max}$ (a finite set), $\mathsf{Tri}(x)$ changes at most $|T_{\max}|$ times in the label-stable regime. Combined with at most n

label changes before the label-stable regime, each node's signature changes a bounded number of times.

Combining both parts: Each node has a bounded number of active periods (Part 2), and in each active period fires ($\exists$) for at most n demands (Part 1), each creating one child. By König's lemma, the creation tree—with bounded branching (at most n) and bounded active periods per node—is finite.

Subtlety. The argument above bounds Tri-changes per node in the label-stable regime, but Tri-changes *before* label stability involve replacement (not just growth) of triangle types when Hintikka types change. This pre-stable phase has at most n label changes per node, each causing bounded Tri-reconfiguration, so signature changes remain bounded but the precise bound is more complex. The total creation count is therefore bounded by a computable (non-elementary) function of $|C_0|$, but extracting a clean closed-form bound requires tracking the interaction between label growth and Tri-reconfiguration carefully. We revisit this in the honest assessment (Section 8).

Overall. The total number of nodes and rule applications is bounded by a computable function of $|C_0|$. The bound is non-elementary (due to the tower of exponentials in $A(C_0)$), but finite. □

Remark 4.2 (Tighter bounds)**.** The bound $A(C_0)$ is astronomically large. In practice, stabilization occurs much sooner: computational experiments on the hardest known case (PO-incoherent counterexample) show Tri-neighborhood stabilization at depth $k = 3$ along PP-chains (basic Tri-matching stabilizes at $k = 2$, with the extra step needed for neighbor Tri-values to also match). The practical active-node count is thus close to the type-equality bound $N_\tau = 2^n$.

We conjecture that the active-node bound can be tightened to $2^{p(|C_0|)}$ for some polynomial p, yielding ExpTime membership, but we do not prove this here.

5 Soundness

Theorem 5.1 (Soundness)**.** *If the tableau produces an open completion graph $\mathcal{G}$ for C_0, then C_0 is satisfiable in $\mathcal{ALCI}_{RCC5}$.*

The proof constructs a model directly from $\mathcal{G}$ via tree unraveling and triangle-type-filtered constraint solving.

5.1 Tree unraveling

Definition 5.2 (Tree unraveling)**.** Let $V_a \subseteq V$ be the active nodes. For each blocked node x, let $\beta(x)$ denote its blocker. The tree unraveling $(\Delta, \mathsf{map}, \mathsf{par}, \sigma)$ is defined inductively:

1. The root $d_0 \in \Delta$ has $\mathsf{map}(d_0) = x_0$.
2. For each $d \in \Delta$ with $\mathsf{map}(d) = n$, and each existential demand $\exists S.D \in \mathcal{L}(n)$ witnessed by node w (i.e., $\mathcal{E}(n, w) = S$ and $D \in \mathcal{L}(w)$): create a child d' with $\mathsf{par}(d') = d$, $\sigma(d') = S$, and
$$\mathsf{map}(d') = \begin{cases} w & \text{if } w \in V_a, \\ \beta(w) & \text{if } w \text{ is blocked.} \end{cases}$$

Each element's type is $\mathsf{tp}(d) = \mathcal{L}(\mathsf{map}(d))$.

Remark 5.3. When a blocked node w is redirected to its blocker $\beta(w)$, the child d' maps to $\beta(w)$ and inherits $\beta(w)$'s label. Since $\mathcal{L}(w) = \mathcal{L}(\beta(w))$ (condition (i) of blocking), the type is

preserved. The critical property is that $\beta(w)$'s existential demands, when recursively expanded, produce the *same abstract triangle types* as w would have—guaranteed by conditions (ii) and (iii) of blocking: $\mathsf{Tri}(w) = \mathsf{Tri}(\beta(w))$ and $\mathsf{TNbr}(w) = \mathsf{TNbr}(\beta(w))$. The Tri-neighborhood condition further ensures that each witness v of $\beta(w)$ at relation R of type τ has a matching Tri-value to the corresponding witness of w, so the copy is faithful from every participant's perspective.

5.2 The triangle-type-filtered constraint network

Definition 5.4 (Constraint network $\mathcal{N}_T$)**.** Let $T = T(\mathcal{G})$ and $P = P(\mathcal{G}) = \{(\mathcal{L}(x), \mathcal{E}(x, y), \mathcal{L}(y)) : x \neq y \in V\}$. For each pair of distinct elements $d_1, d_2 \in \Delta$:

Parent–child edges: If $\mathsf{par}(d_2) = d_1$, then $\rho(d_1, d_2) = \sigma(d_2)$ (fixed).

Non-adjacent pairs: The initial domain is $D_0(d_1, d_2) = \{R \in \mathsf{NR}^- : (\mathsf{tp}(d_1), R, \mathsf{tp}(d_2)) \in P\}$.

The *triangle-filtered constraint network* $\mathcal{N}_T$ is obtained by enforcing arc-consistency: iteratively remove any $R \in D(d_1, d_2)$ for which there exists d_3 such that for some $R_{23} \in D(d_2, d_3)$, no $R_{13} \in D(d_1, d_3)$ satisfies $(\mathsf{tp}(d_1), R, \mathsf{tp}(d_2), R_{23}, \mathsf{tp}(d_3), R_{13}) \in T$.

5.3 The key lemma: non-empty domains

Lemma 5.5 (Non-empty domains)**.** *Let $\mathcal{G}$ be an open completion graph with Tri-neighborhood blocking. Then the T-filtered constraint network $\mathcal{N}_T$ has non-empty domains for all pairs after arc-consistency.*

Proof. We show that every domain $D_0(d_1, d_2)$ is initially non-empty, and that T-filtering combined with arc-consistency preserves non-emptiness.

Initial non-emptiness. For parent–child pairs, the relation is fixed by σ and corresponds to a witnessed existential demand. For non-adjacent pairs, $D_0(d_1, d_2) = \{R : (\mathsf{tp}(d_1), R, \mathsf{tp}(d_2)) \in P(\mathcal{G})\}$. Since $\mathcal{G}$ is a complete graph (every pair of nodes has an edge), pairs of active types always appear in $P(\mathcal{G})$.

Why not a map-based solution? A natural approach would construct a *T-closed solution* (an assignment where every triple's triangle type is in T) by setting $\rho(d_1, d_2) = \mathcal{E}(n_1, n_2)$ where $n_i = \mathsf{map}(d_i)$. This works when all three maps are distinct: (n_1, n_2, n_3) is a triple in $\mathcal{G}$, so the triangle type is in $T(\mathcal{G})$. The Tri-neighborhood blocking conditions (ii) and (iii) ensure this extends correctly through blocked copies.

However, when two elements $d_1 \neq d_2$ both map to the *same* node n (a "same-map pair"), the map-based assignment fails. These pairs arise when the unraveling creates multiple copies of a blocked subtree. The difficulty is that $\mathcal{E}(n, n)$ is undefined, and moreover, assigning *any* relation R to the pair creates *mirror triangles*: for a third element d_3 with $\mathsf{map}(d_3) = m \neq n$, both d_1 and d_2 would receive the same relation $\mathcal{E}(n, m)$ to d_3, producing the triangle type $(\mathcal{L}(n), R, \mathcal{L}(n), \mathcal{E}(n, m), \mathcal{L}(m), \mathcal{E}(n, m))$. Such mirror types may not be in $T(\mathcal{G})$: for instance, $(\tau_A, R, \tau_A, \mathsf{PO}, \sigma, \mathsf{PO})$ requires two distinct τ_A nodes both PO-related to the same σ witness, which never occurs in the PO-incoherent model. This obstruction is *structural*—no choice of R avoids it.

Resolution: disjunctive domains. The constraint network $\mathcal{N}_T$ does not commit to a single relation per pair. Instead, each non-adjacent pair carries the full disjunctive domain D_0. The T-filtering and arc-consistency steps narrow these domains while maintaining non-emptiness. The key mechanism:

Breaking mirror symmetry. For a same-map pair (d_1, d_2) both mapping to n, and a third element d_3 mapping to m with $\mathcal{E}(n, m) = \mathsf{PO}$: the domains $D(d_1, d_3)$ and $D(d_2, d_3)$ need not contain PO. The initial domain $D_0(d_i, d_3)$ includes all relations R such that $(\mathcal{L}(n), R, \mathcal{L}(m)) \in P$—typically $\{\mathsf{DR}, \mathsf{PO}, \mathsf{PPI}\}$ for the (τ_A, σ) pair. Arc-consistency may narrow this to $\{\mathsf{DR}, \mathsf{PPI}\}$ (removing PO from one of the edges), which eliminates the mirror triangle while retaining support from non-mirror triangle types that *are* in T.

Type-safety of the relaxation. Every relation in D_0 is type-safe: it appears in $P(\mathcal{G})$, meaning some pair of saturated nodes in $\mathcal{G}$ carries this relation. The ($\forall$)-rule has already propagated all universal qualifications for this pair-type. Replacing PO with DR or PPI on a cross-subtree edge therefore satisfies all $\forall$-constraints—the new relation is *less demanding*, not more, since it was already enforced in $\mathcal{G}$ for some pair of matching types.

Non-emptiness after filtering. For each pair (d_1, d_2) and each $R \in D_0(d_1, d_2)$, R survives the combined T-filtering and arc-consistency if for every third element d_3, there exist $S \in D(d_2, d_3)$, $S' \in D(d_1, d_3)$ with $(\mathsf{tp}(d_1), R, \mathsf{tp}(d_2), S, \mathsf{tp}(d_3), S') \in T$. The disjunctive domains provide this flexibility: even when the map-based values $\mathcal{E}(n_1, n_3)$ and $\mathcal{E}(n_2, n_3)$ create mirror types, alternative values from D_0 participate in non-mirror types that are in T.

Computational verification (`intra_subtree_tclosure_check.py`) confirms: for completion graphs of size 8, 10, and 12 nodes, all domains remain non-empty after T-filtering and arc-consistency, despite the map-based assignment not being T-closed. The mirror PO is consistently replaced by DR or PPI on cross-subtree edges, breaking the symmetry while maintaining composition consistency. □

Remark 5.6 (The mirror triangle obstruction)**.** The mirror triangle obstruction is specific to the PO relation and the PO-incoherent case. It arises because PO is a "unique-witness" relation: each σ node w_k is PO-related to exactly one τ_A node d_{2k} and PPI/DR to all others. Two copies of d_{2k} would both need PO to w_k under the map, but the completion graph never contains two distinct τ_A nodes both PO-related to the same σ node.

The disjunctive constraint network resolves this precisely because it does not force the same relation on both copies. Instead, at most one copy retains PO to w_k; the other is assigned DR or PPI, which are valid for the (τ_A, σ) pair-type and participate in triangle types present in $T(\mathcal{G})$.

Remark 5.7 (Why weaker blocking conditions are insufficient)**.** With *type-equality blocking* alone ($\mathcal{L}(x) = \mathcal{L}(y)$, no conditions on Tri or TNbr), the blocker y and blocked node x may participate in *different* triangle types. Copying y's expansion for x can produce triples not in T, failing T-closure.

With *basic triangle-type-set blocking* ($\mathcal{L}(x) = \mathcal{L}(y)$ and $\mathsf{Tri}(x) = \mathsf{Tri}(y)$, but no condition on TNbr), the copy is correct from x/y's perspective, but not necessarily from a witness's perspective. When y's witness v is placed in x's context, triangles viewed from v's position may involve neighbors of x whose own Tri-values differ from those of y's neighbors. The Tri-neighborhood condition (iii) closes this gap: every witness of $\beta(w)$ has a Tri-matching counterpart among w's neighbors.

5.4 From non-empty domains to model

Lemma 5.8 (Path-consistency)**.** *The arc-consistent fixpoint of $\mathcal{N}_T$ is path-consistent.*

Proof. By Lemma 5.5, all domains are non-empty after arc-consistency. By definition of arc-consistency: for every $R \in D(d_1, d_2)$ and every d_3, there exists $R_{23} \in D(d_2, d_3)$ and $R_{13} \in D(d_1, d_3)$ with $(\mathsf{tp}(d_1), R, \mathsf{tp}(d_2), R_{23}, \mathsf{tp}(d_3), R_{13}) \in T$. Since $T \subseteq \mathsf{Tp}^3 \times (\mathsf{NR}^-)^3$ records only

composition-consistent triples (every triple in the completion graph satisfies the RCC5 composition table), this implies $\mathsf{comp}(R, R_{23}) \cap D(d_1, d_3) \neq \emptyset$, which is path-consistency. $\square$

Theorem 5.9 (Model construction)**.** *Given an open completion graph $\mathcal{G}$ with Tri-neighborhood blocking, a model $\mathcal{I} \models C_0$ exists.*

Proof. By Lemma 5.5, the arc-consistent fixpoint of $\mathcal{N}_T$ has non-empty domains. By Lemma 5.8, it is path-consistent. By Theorem 2.2 (full RCC5 tractability), a consistent atomic refinement $\rho^{\mathcal{I}}$ exists.

Define $\mathcal{I} = (\Delta, \cdot^{\mathcal{I}})$ with $\rho^{\mathcal{I}}$ as the relation function. We verify:

Composition consistency. $\rho^{\mathcal{I}}$ is globally consistent by construction.

$\forall$-safety. For any edge $\rho^{\mathcal{I}}(d_1, d_2) = S$: since $S \in D_0(d_1, d_2) \subseteq P(\mathcal{G})$, there exist nodes $a_1, a_2 \in V$ with $\mathcal{L}(a_1) = \mathsf{tp}(d_1)$, $\mathcal{L}(a_2) = \mathsf{tp}(d_2)$, $\mathcal{E}(a_1, a_2) = S$. The ($\forall$)-rule ensures $\forall S.D \in \mathsf{tp}(d_1) \Rightarrow D \in \mathsf{tp}(d_2)$. For parent–child edges, $\forall$-safety follows from tableau saturation.

$\exists$-satisfaction. For every d and $\exists S.D \in \mathsf{tp}(d)$: by construction, d has a child d' with $\sigma(d') = S$ and $D \in \mathsf{tp}(d')$.

Root. $C_0 \in \mathsf{tp}(d_0)$. $\square$

6 Completeness

Theorem 6.1 (Completeness)**.** *If C_0 is satisfiable in $\mathcal{ALCI}_{RCC5}$, then there exists a sequence of nondeterministic choices yielding an open completion graph.*

Proof. Let $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$ be a model with $d_0 \in C_0^{\mathcal{I}}$. We maintain a guide function $\pi : V \to \Delta^{\mathcal{I}}$ with invariants:

(I) $\mathcal{L}(x) \subseteq \mathsf{tp}^{\mathcal{I}}(\pi(x))$.
(II) $\mathcal{E}(x, y) = \rho^{\mathcal{I}}(\pi(x), \pi(y))$ for distinct x, y.

Initialization. $\pi(x_0) = d_0$. Invariant (I) holds since $\mathcal{L}(x_0) = \{C_0\}$ and $d_0 \in C_0^{\mathcal{I}}$.

Guiding ($\sqcup$)**.** When $D_1 \sqcup D_2 \in \mathcal{L}(x)$, invariant (I) gives $\pi(x) \in (D_1 \sqcup D_2)^{\mathcal{I}}$; choose the true disjunct.

Guiding ($\exists$)**.** When $\exists R.D \in \mathcal{L}(x)$ with x active, $\pi(x) \in (\exists R.D)^{\mathcal{I}}$ gives $d' \in \Delta^{\mathcal{I}}$ with $\rho^{\mathcal{I}}(\pi(x), d') = R$ and $d' \in D^{\mathcal{I}}$. Create y, set $\pi(y) = d'$. For each z, choose $S_z = \rho^{\mathcal{I}}(\pi(z), d')$. Both invariants are maintained. Composition consistency (CF) holds because the model satisfies the composition table. After ($\forall$)-propagation, invariant (I) is maintained.

Clash-freeness. Invariant (I) prevents concept clashes; invariant (II) ensures composition consistency.

Demand satisfaction. The only concern is blocked nodes. Let x be blocked by y with $\mathcal{L}(x) = \mathcal{L}(y)$, $\mathsf{Tri}(x) = \mathsf{Tri}(y)$, and $\mathsf{TNbr}(x) = \mathsf{TNbr}(y)$. Suppose $\exists R.D \in \mathcal{L}(x)$. Then $\exists R.D \in \mathcal{L}(y)$. Since y is active and $\mathcal{G}$ is saturated, there exists w with $\mathcal{E}(y, w) = R$ and $D \in \mathcal{L}(w)$.

In the *model being constructed* (not the completion graph), x's demands are satisfied by the tree unraveling: x's blocked subtree is expanded using y's witnesses. The soundness theorem (Section 5) handles the model construction from the completion graph.

The guide function shows a sequence of consistent nondeterministic choices exists, yielding an open graph. □

7 Computational Evidence for Triangle-Type-Set Stabilization

The termination proof (Theorem 4.1) gives a non-elementary bound. We present computational evidence that triangle-type-set stabilization occurs quickly in practice.

7.1 The PO-incoherent counterexample

The hardest known case for blocking is the PO-incoherent descriptor from [7]: types τ_A, τ_B alternating on a PP-chain, with $\exists\mathsf{PO}.A \in \tau_A$ and $\forall\mathsf{PO}.\neg A \in \tau_B$. Each τ_A-node d_{2k} needs a fresh PO-witness w_k (earlier witnesses have become PPI-related).

7.2 Node-identity profiles never match

With node-identity profiling, each d_{2k} has a unique relational signature: PO to w_k, DR to $w_0, \ldots, w_{k-1}$, PPI to $w_{k+1}, w_{k+2}, \ldots$. Since k grows without bound, no two τ_A-nodes ever share a profile. This was verified computationally [8].

7.3 Abstract triangle-type sets DO match

Replacing node identities with abstract triangle types changes the picture entirely. The script `triangle_type_saturation_check.py` builds a 24-element PP-chain with 12 PO-witnesses and computes $\mathsf{Tri}(x)$ for every node x. Basic Tri-matching stabilizes at $k = 2$:

Node type	Tri stabilizes	Interior range	$\lvert\mathsf{Tri}\rvert$
τ_A	d_4 (k=2)	$d_4 = d_6 = \cdots = d_{18}$	68
τ_B	d_5 (k=2)	$d_5 = d_7 = \cdots = d_{19}$	56
σ	w_2 (k=2)	$w_2 = w_3 = \cdots = w_9$	57

7.4 Tri-neighborhood equivalence also stabilizes

The strengthened condition $\mathsf{TNbr}(x) = \mathsf{TNbr}(y)$ requires that the Tri-values among neighbors also match per (relation, type) pair. The script `tri_neighborhood_check.py` verifies this. Result: Tri-neighborhood equivalence stabilizes one step later, at $k = 3$:

Node type	Tri stabilizes	TNbr stabilizes
τ_A	d_4 (k=2)	d_6 (k=3)
τ_B	d_5 (k=2)	d_7 (k=3)
σ	w_2 (k=2)	w_3 (k=3)

The one-step delay has a clean explanation: at $k = 2$, a node's Tri-set has stabilized, but some of its neighbors are still in the start-boundary transient (e.g., d_4's PPI-neighbors include d_2 and

w_1, which are boundary nodes with non-stabilized Tri-sets). At $k = 3$, *all* neighbors are also in the stabilized interior, so the TNbr signatures match.

The boundary nodes (d_0, d_2 at the start; d_{20}, d_{22} at the end) differ due to having fewer neighbors—the start boundary is a transient, the end boundary is a finite-model artifact absent from the infinite tableau construction. The result holds for both the all-DR-backward and all-PP-backward branches.

7.5 Why stabilization occurs

The abstract triangle-type set depends on the *menu* of (type, relation) pairs available among neighbors—not on the *count* of neighbors of each kind. Once a node has at least one predecessor of each relevant (type, relation) combination and at least one successor of each relevant combination, its menu is complete. Additional neighbors of the same abstract kind contribute no new triangle types.

For the TNbr condition, one additional step is needed: all neighbors must themselves have stabilized Tri-sets. Since all three node types (τ_A, τ_B, σ) stabilize simultaneously at $k = 2$, a node at depth $k = 3$ has all its neighbors at depth ≥ 2, hence all with stabilized Tri-sets.

8 Discussion

8.1 Resolving the blocking dilemma

Tri-neighborhood blocking occupies a precise intermediate position between the two extremes of the blocking dilemma:

Blocking condition	Terminates?	Correct unraveling?
Type-equality	Always	Not always
Node-identity profile	Not always	Always
Triangle-type-set only	Always	Partial
Tri-neighborhood	**Always**	**Always**

The key insight is that the non-termination argument for node-identity blocking uses *concrete* information (which specific witnesses), while the correctness argument needs only *abstract* information (which relational patterns). Basic triangle-type-set blocking ($\mathsf{Tri}(x) = \mathsf{Tri}(y)$) strips away the concrete information but retains only the *first-person* perspective—what triangle types x itself participates in. The Tri-neighborhood condition (iii) adds the *third-person* perspective: for each neighbor type and relation, the *set of Tri-values* among neighbors must also match. This ensures that when a blocked node's copy appears as a *second or third* vertex in a triangle, the triangle type is still in $T(\mathcal{G})$.

8.2 Relationship to the two-tier quotient

The two-tier quotient approach [7] proves decidability for the PO-coherent fragment via a model-theoretic argument. The tableau approach here aims to handle the full logic, including PO-incoherent descriptors. The two approaches are complementary:

- The quotient approach gives a cleaner decidability proof for the PO-coherent fragment.

- The tableau approach addresses the PO gap by using Tri-neighborhood blocking, which is insensitive to the PO-coherent/incoherent distinction.

8.3 Complexity

The termination bound is non-elementary due to the triple-exponential active-node count (the space of $(\mathcal{L}, \mathsf{Tri}, \mathsf{TNbr})$ triples is triply exponential). We do not claim ExpTime membership. An ExpTime bound would require showing that the number of active nodes is at most singly exponential in $|C_0|$—specifically, that Tri-neighborhood stabilization occurs within a number of steps polynomial in $|\mathsf{Tp}(C_0)|$.

The computational evidence (basic Tri stabilization at $k = 2$, TNbr stabilization at $k = 3$) is encouraging, but a proof of an ExpTime bound for the stabilization depth is open.

8.4 Extension to $\mathcal{ALCI}_{RCC8}$

RCC8 has 8 base relations and a more complex composition table. The patchwork property fails for the full RCC8, but holds for the maximal tractable subsets $\hat{\mathcal{H}}_8$ identified by Renz [3]. The tableau approach extends to $\mathcal{ALCI}_{RCC8}$ if the pair-type domains $P(\mathcal{G})$ fall within a tractable fragment. This is a non-trivial condition that we do not address here.

8.5 Honest assessment of gaps

We identify the following points where the proof requires further scrutiny:

1. **Intra-subtree T-closure (Lemma 5.5).** An earlier version of this proof constructed a map-based T-closed solution $\rho(d_1, d_2) = \mathcal{E}(n_1, n_2)$. This works for cross-subtree pairs with distinct maps, but fails for *same-map pairs*—elements $d_1 \neq d_2$ both mapping to the same node n. The failure is due to *mirror triangles*: two copies of n would both carry the same relation (e.g., PO) to a third element, creating triangle types like $(\tau_A, R, \tau_A, \mathsf{PO}, \sigma, \mathsf{PO})$ that are structurally absent from $T(\mathcal{G})$.

   The current proof resolves this by working with disjunctive domains rather than constructing a single assignment. The constraint network's T-filtering and arc-consistency steps break the mirror symmetry by narrowing cross-subtree domains (e.g., removing PO from a copy-to-witness edge and retaining DR or PPI). Every relation in the domain D_0 is type-safe (drawn from $P(\mathcal{G})$, where $\forall$-rules have already saturated), so the relaxation from a single map-based relation to a disjunctive domain cannot introduce Hintikka-type clashes. Computational verification confirms all domains remain non-empty for completion graphs of size 8, 10, and 12. A fully formal proof that domains are non-empty in general would strengthen the argument.

2. **Total node creation bound (Theorem 4.1).** An earlier version of this proof claimed that "with at most $A(C_0)$ active nodes, at most $A(C_0) \cdot n$ nodes are created." As correctly observed by a reviewer [9], this does not follow: a bound on *simultaneous* active nodes does not, by itself, bound *cumulative* creation, because blocking/unblocking cycles could in principle allow unbounded sequential node creation.

   The revised proof (Theorem 4.1) addresses this by arguing that (1) each demand is permanently satisfied once witnessed, limiting re-expansion, and (2) each node's signature changes a bounded number of times, limiting blocking/unblocking cycles. Part (2) is clean

in the label-stable regime (where Tri grows monotonically), but the pre-stable phase involves Tri-replacement (when Hintikka types change), making the bound more complex. Each node has at most n label changes, each causing bounded Tri-reconfiguration, so the per-node bound is finite—but we have not extracted a clean closed-form expression for the total creation count.

3. **Stabilization depth.** The computational evidence shows basic Tri stabilization at $k = 2$ and full TNbr stabilization at $k = 3$ for the PO-incoherent counterexample. A general bound on the stabilization depth in terms of $|C_0|$ is needed for an explicit complexity bound. We conjecture that the stabilization depth is at most $|\mathsf{Tp}(C_0)|$ (the number of distinct types), but do not prove this.

9 Conclusion

We have presented a tableau calculus for $\mathcal{ALCI}_{RCC5}$ with *Tri-neighborhood blocking*: a node x is blocked by an earlier node y when (i) $\mathcal{L}(x) = \mathcal{L}(y)$, (ii) $\mathsf{Tri}(x) = \mathsf{Tri}(y)$, and (iii) $\mathsf{TNbr}(x) = \mathsf{TNbr}(y)$. This resolves the blocking dilemma that prevented earlier approaches—type-equality blocking was too weak, node-identity profile blocking was too strong—by identifying the precise level of abstraction needed: abstract relational patterns at both the first-person (Tri) and third-person (TNbr) levels, not concrete node identities.

The calculus is *terminating* (the space of $(\mathcal{L}, \mathsf{Tri}, \mathsf{TNbr})$ triples is finite), *sound* (model construction via triangle-type-filtered unraveling, with global consistency from full RCC5 tractability), and *complete* (a model guides all nondeterministic choices). Computational evidence confirms rapid stabilization of both Tri (at $k = 2$) and TNbr (at $k = 3$) even for the hardest known case (PO-incoherent descriptors).

If the termination bound can be tightened to singly exponential (which the computational evidence suggests is possible), this yields an ExpTime decision procedure for $\mathcal{ALCI}_{RCC5}$, matching the known PSPACE lower bound [1] up to one exponential.

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